

14/14

100%

Name: _____

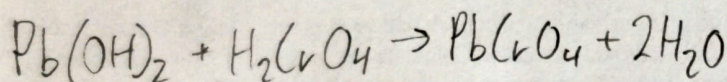
These questions assess your understanding of the material from Chapters 3, 4 of Openstax Chemistry. Please write your name on the sheets of paper that you use. Carefully read each question. For questions with numerical calculations, you must show each major step necessary to find your final answer. For questions without numerical calculations, you must include clear and correct reasoning to support your answer. Remember to report the units and correct number of significant figures in your answers.

1. Ryan has a 4.00-M solution of chromium (III) cyanide ($\text{Cr}(\text{CN})_3$). If he pours 1.58 mL of chromium (III) cyanide into a reaction flask, what is the mass of chromium (III) cyanide that he has added to the flask? Be sure your answer has the correct number of significant figures.

ANSWER: 0.822 g $\text{Cr}(\text{CN})_3$ ✓

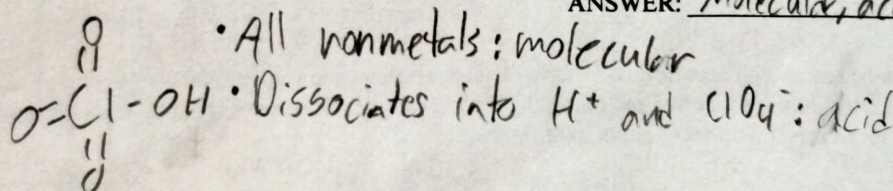
$$\frac{4.00 \text{ mol}}{\text{L}} \times \frac{1 \text{ L}}{1000 \text{ mL}} \times 1.58 \text{ mL} \times \frac{130.05 \text{ g}}{1 \text{ mol}} = 0.821916 \text{ g } \text{Cr}(\text{CN})_3 \quad 3 \text{ SF}$$

2. Lead (II) hydroxide reacts with chromic acid to form lead (II) chromate and water. Calculate the moles of lead (II) hydroxide needed to produce 1.793 moles of water. Ensure your answer has the correct number of significant figures.

ANSWER: 0.8965 mol $\text{Pb}(\text{OH})_2$ ✓1 $\text{Pb}(\text{OH})_2$ makes 2 H_2O

$$\frac{1.793 \text{ mol } \text{H}_2\text{O}}{2 \text{ mol } \text{H}_2\text{O} / \text{mol } \text{Pb}(\text{OH})_2} = 0.8965 \text{ mol}$$

3. Determine which of the following labels apply to the chemical compound HClO_4 : ionic, molecular, acid, and base.

ANSWER: Molecular, acid ✓

4. A particular planet has a mass of $3.3 \times 10^{27} \text{ g}$ and a coin has a mass of $1.303 \times 10^0 \text{ g}$. How many moles of this particular planet have a mass equal to the mass of a coin? Ensure your answer has the correct number of significant figures.

ANSWER: $6.6 \times 10^{-52} \text{ mol planet}$ ✓

$$\frac{1.303 \text{ g}}{3.3 \times 10^{27} \text{ g}} = 3.94848 \times 10^{-28} \frac{\text{Planets}}{\text{Coin}} \quad 2 \text{ SF, odd rounds up}$$

$$\frac{3.94848 \times 10^{-28} \text{ Planets (atoms)}}{6.022 \times 10^{23} \text{ atoms/mol}} = 6.5567666 \times 10^{-52} \frac{\text{mol Planets}}{\text{Coin}}$$

5. David prepares a solution of zinc cyanide ($\text{Zn}(\text{CN})_2$) by measuring out 1.329 g of zinc cyanide into a 50.-mL volumetric flask and filling the flask to the mark with water. Calculate the concentration of the solution in mol/L. Be sure your answer has the correct number of significant figures.

ANSWER: 0.23 M $\text{Zn}(\text{CN})_2$ soln. ✓

$$\frac{1.329 \text{ g}}{117.44 \text{ g/mol}} = 0.011316 \text{ mol}$$

$$\frac{0.011316 \text{ mol}}{0.050 \text{ L}} = 0.22632 \text{ M}$$

2 SF

+ 5

6. Tyler prepares a solution of ammonium iodide (NH_4I) by measuring out 36.05 mmol of ammonium iodide into a 500.-mL volumetric flask and filling the flask to the mark with water. Calculate the concentration of the solution in mol/L. Be sure your answer has the correct number of significant figures.

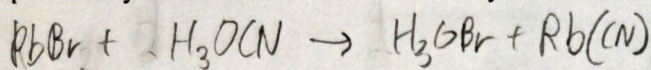
ANSWER: 0.0721 M NH_4I soln.
3 SF

$$\frac{0.03605 \text{ mol}}{0.500 \text{ L}} = 0.0721 \text{ M}$$

↑
3 SF

7. Rubidium bromide reacts with hydronium cyanide to form hydronium bromide and rubidium cyanide. Suppose 14.9 g of rubidium bromide is mixed with 3.07 g of hydronium cyanide. Calculate the maximum mass of hydronium bromide that could be produced by the chemical reaction. Ensure that your answer has the correct number of significant figures.

ANSWER: 6.74 g H_3OBr
3 SF

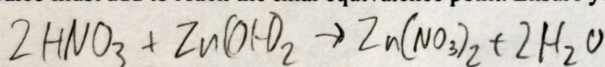


$$\frac{14.9 \text{ g}}{165.37 \text{ g/mol}} = 0.0901 \text{ mol} \quad \frac{3.07 \text{ g}}{45.044 \text{ g/mol}} = 0.0681556 \text{ mol } \text{H}_3\text{OCN}$$

↑ limiting reagent

8. Caleb weighs out 1.71 g of nitric acid (HNO_3) into a 50.-mL volumetric flask and dilutes the solution to the mark with distilled water. He plans to titrate the acid with 0.389-M zinc hydroxide solution. Calculate the volume of zinc hydroxide solution that Caleb must add to reach the final equivalence point. Ensure your answer has the correct number of significant figures.

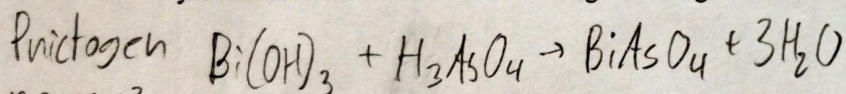
ANSWER: 34.9 mL Zn(OH)_2 soln.
3 SF



$$\frac{1.71 \text{ g } \text{HNO}_3}{63.018 \text{ g/mol}} = 0.027135 \text{ mol } \text{HNO}_3 = \frac{0.013567 \text{ mol } \text{Zn(OH)}_2}{0.389 \text{ mol/L}} = 0.03488 \text{ L } \text{Zn(OH)}_2 \text{ soln.}$$

9. Bismuth hydroxide reacts with arsenic acid to form bismuth arsenate and water. Suppose 59.94 g of bismuth hydroxide is mixed with 52.51 g of arsenic acid. Calculate and report the theoretical yield of water formed by this chemical reaction. Ensure that your answer has the correct number of significant figures.

ANSWER: 12.46 g H_2O
4 SF



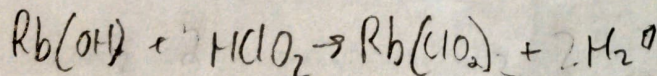
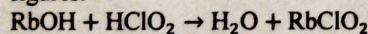
$$\frac{59.94 \text{ g}}{260 \text{ g/mol}} = 0.230535 \text{ mol}$$

↑ limiting

$$\frac{52.51 \text{ g}}{141.956 \text{ g/mol}} = 0.36993 \text{ mol}$$

$$\frac{0.230535 \text{ mol } \text{Bi(OH)}_3}{1 \text{ mol } \text{Bi(OH)}_3} \times (3 \text{ mol } \text{H}_2\text{O}) = 12.46 \text{ g}$$

10. Given this unbalanced chemical equation, calculate and report the percent yield of RbClO_2 if 11.69 g of RbClO_2 is produced from the reaction of 18.08 g of RbOH and 7.655 g of HClO_2 . Ensure that your answer has the correct number of significant figures.



ANSWER: 68.36% yield
4 SF

$$\frac{18.08 \text{ g}}{102.4758 \text{ g/mol}} = 0.176432 \text{ mol}$$

$$\frac{7.655 \text{ g}}{63.458 \text{ g/mol}} = 0.11182 \text{ mol } \text{HClO}_2 = 0.11182 \text{ mol } \text{RbClO}_2$$

$$\frac{11.69 \text{ g}}{17.0944 \text{ g/mol}} \cdot 100\% \approx 68.36\%$$

11. Katherine has a 0.2161-M solution of ammonium oxide ((NH₄)₂O). If she needs to add 1.96 g of ammonium oxide to a reaction flask, what volume of the solution should she pour? Be sure your answer has the correct number of significant figures.

ANSWER: 174 mL
3 SF

$$\frac{1.96 \text{ g}}{52.084 \text{ g/mol}} = 0.037632 \text{ mol} = \frac{0.037632 \text{ mol}}{0.2161 \text{ mol/L}} = 0.17414 \text{ L}$$

12. As part of his routine quality-control checks, Justin, an analytical chemist at Southern Union Superb Chemicals Corporation, must determine the concentration of a 2000-L fresh batch of sulfuric acid solution. He takes a 25-mL sample of sulfuric acid solution and titrates it with a manganese (II) hydroxide solution, which he made by dissolving 7.60 g of solid manganese (II) hydroxide in 1.375 L of distilled water. If Justin finds that it takes 406 mL of the manganese (II) hydroxide solution for the titration to reach the equivalence point, what molarity of the sulfuric acid solution does Justin calculate?

ANSWER: 1.009 M H₂SO₄ (M)
2 SF

$$\frac{7.60 \text{ g}}{88.954 \text{ g/mol}} = 0.08544 \text{ mol Mn(OH)}_2 = \frac{0.08544 \text{ mol}}{1.375 \text{ L}} = 0.06214 \text{ mol/L} \cdot 0.406 \text{ L} = 0.02523 \text{ mol}$$

$$\frac{0.02523 \text{ mol}}{0.025 \text{ L}} = 1.009 \text{ M}$$

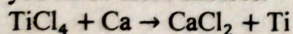
$\text{Mn(OH)}_2 + \text{H}_2\text{SO}_4 \rightarrow \text{MnSO}_4 + 2\text{H}_2\text{O}$

13. Julia needs to make 225 mL of aqueous potassium chlorate (KClO₃) working solution. She plans to do this by adding distilled water to a 87.5-mL stock solution of 0.078-M aqueous potassium chlorate. What will be the concentration of her working solution? Report your answer in mM. Be sure your answer has the correct number of significant figures.

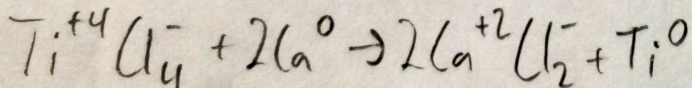
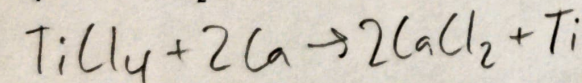
ANSWER: 30. mM
2 SF

$$0.0875 \text{ L} \cdot \frac{0.078 \text{ mol}}{\text{L}} = 0.006825 \text{ mol} = \frac{0.006825 \text{ mol}}{0.225 \text{ L}} = 0.03033 \text{ M} \text{ or } 30.33 \text{ mM}$$

14. Determine whether the following reaction is an oxidation-reduction (redox) reaction. If the reaction is a redox reaction, write down the formula of the reactant that is oxidized and the formula of the reactant that is reduced. Provide an explanation for why you made these choices.



ANSWER: Yes



Yes, Ti is reduced and Ca is oxidized

$\text{Ti}^{+4} \xrightarrow{\text{red}} \text{Ti}^0$ (TiCl₄ oxidizing agent)
 $\text{Ca}^0 \xrightarrow{\text{ox}} \text{Ca}^{+2}$ (Ca is reducing agent)

CHM111 Exam 2 on chapter(s): 3, 4

1. 0.822 g
2. 0.8965 moles
3. molecular, acid
4. 6.6×10^{-52} moles
5. $0.23 \frac{\text{mol}}{\text{L}}$
6. $0.0721 \frac{\text{mol}}{\text{L}}$
7. 6.75 g
8. 34.9 mL
9. 12.46 g
10. 68.36%
11. 0.174 L
12. 1.0 M
13. 30. mM
14. Oxidized: Ca Reduced: Ti